Problem 07: Personalized Financial Risk Intelligence and Early-Warning System for Small Businesses

Domain: FinTech | Time-Series ML | Anomaly Detection | Explainable AI
Difficulty: Medium-Hard to Hard

Problem

Small businesses face cash-flow trouble from late payments, shifting expenses, seasonality and uncertain revenue. Accounting dashboards show the past, not emerging liquidity problems.

Objective

Build a forecasting and decision-support system that forecasts cash flow, flags unusual transactions, estimates short-term liquidity risk and recommends actions via scenarios.

Users

Small-business owners, accountants, lenders.

Core Requirements

- Import revenue, expense, invoice and payment records.
- Clean data; flag missing/inconsistent records.
- Forecast inflows and outflows.
- Detect unusual spending/payment patterns.
- Estimate short-term cash-deficit risk with a clearly defined target (e.g., balance < 0 in next 30 days).
- Identify drivers of risk.
- Simulate delayed payments, higher expenses, lower revenue.
- Recommend actions (chase overdue invoices, review discretionary spend).
- Show forecast ranges, not only point estimates.
- Dashboard for trends, risk and scenarios.

Advanced Challenges

- Probabilistic forecasting (quantiles).
- Combining transaction anomalies with aggregate forecasts.
- Seasonality and regime change.
- Risk evaluation under economic scenarios.
- Explanations consistent with numbers.
- Stopping the LLM from inventing financial facts.

Suggested Datasets

Primarily a documented synthetic generator (seasonality, payment delays, shocks); supplement with public invoice/payment datasets on Kaggle and ULB credit-card data for anomaly testing.

Suggested Tech

Pandas, LightGBM/XGBoost with quantile loss, Prophet/statsmodels/NeuralForecast, conformal prediction, SHAP, FastAPI, Plotly, Streamlit.

Reference Architecture

Ingest → validation → feature builder (lags, due dates, aging buckets) → forecaster → liquidity classifier → anomaly detector → scenario engine → recommender with template-grounded text (LLM only rewrites verified numbers).

Evaluation

MAE/RMSE/MAPE, pinball loss and interval coverage, anomaly precision, calibration (Brier/ECE) of risk, backtest with rolling origin, usefulness of recommendations (simulated cash improvement).

Safety Note

Synthetic or authorized data only. Decision support, not financial advice.

Pitfalls
Random-split leakage in time series; point forecasts only; LLM paraphrasing numbers incorrectly.
Suggested Timeline (48h)
0-8h data generator + cleaning; 8-20h forecasting; 20-30h risk + anomalies; 30-40h scenarios + recommendations; 40-48h dashboard.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work